

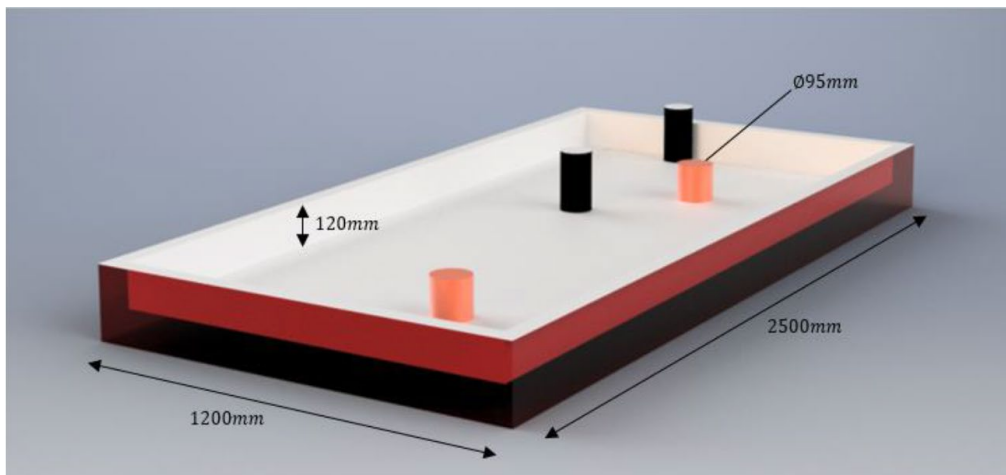
Project 2 – Behavior Control Robotics

The challenge:

- The robot will operate on the surface of a walled table (as depicted in the figure below), starting from an unknown position and orientation.
- The robot autonomously navigates the environment to locate and extinguish two fires as quickly as possible while avoiding collisions with fires, obstacles (quantity unknown), and the table's four walls.
- Most obstacles are static, but 1–2 may move unpredictably during the task.
- The robot must cease all movement immediately after extinguishing the second fire.

The constraints:

- All fires and obstacles are cylindrical with a 10 cm diameter and roughly 12 cm height, positioned randomly across the table surface.
- A fire (comprising an LED-thermistor assembly mounted on the cylinder) can be extinguished by activating a fan for up to 10 seconds. The thermistor automatically deactivates the LED once extinguished. If the robot extinguishes the fire in under 10 seconds, it may proceed immediately.
- A fire can only be extinguished when the robot enters within a 20 cm radius of the fire's centre (measured from the robot's centre point).



The design kit:

A design kit is provided, containing the following components:

- 1x Mobile base (incl. Arduino Mega, shield, battery, 4 motors)
- 1x Small Servo
- 4x 2D120X or 2Y0A41SK (IR Sensor, medium range): Range 40-300mm
- 4x 2Y0A21 (IR Sensor, long range): Range 100-800mm
- 1x BNO 08X motion sensor (used as a single axis gyro)
- 1x HC-SR04 (Ultrasonic Sensor)
- 4x Phototransistors
- 1x Fan
- 1x USB Cable
- 1x Wireless Module (incl 2xHC12, 1xYP-05, 1 USB A to USB mini cable)
- 1x MOSFET (FQP30N06)
- 1x 5V regulator (7805 5V 1A TO-220)
- 1 x Lipo battery (with a smart charger)
- (+ mounting brackets etc available in lab)
- (+ self-made 3D printed brackets are allowed)

Note that only 2 IR sensors of each type may be used in the design.

The demonstration:

- Each robot is allocated 5 minutes for two runs, with the better result used for assessment.
- Two self-powered lights will be used for the demonstration.
- The light will be turned off manually at 10 sec after the fan is turned on, if the light has not gone off earlier, or if an externally powered lights are used for demonstration.
- An unknown number of objects will be randomly placed on the table, and two of them will move toward the robot at separate times.
- Objects may be placed touching each other or against the wall.
- The setup for demonstration will be same for all groups.

The demonstration accounts for 60% of the project assessment, which will be marked according to the following rubrics.

Firefighting (20%)	Obstacle avoidance (20%)	Time (20%)
2 fires extinguished, 10 points	No collision, 10 points	60 sec or less, 10 points
1 fire extinguished, 1 detected, 9 points	1 collision, 9 points	60 to 70 sec, 9 points
1 fire extinguished, 8 points	2 collisions, 8 points	70 to 80 sec, 8 points
1 fire detected, not extinguished, 7 points	3 collisions, 7 points	80 to 90 sec, 7 points
0 fire detected, 6 points	4 collisions, 6 points	90 to 100 sec, 6 points
Base point, 4 points	5 collisions, 5 points	100 sec or above, 5 points
	Base point, 4 points	Base point, 4 points

Note: when the demo is marked, Time will be given Base Points if the robot fails completely in firefighting and obstacle avoidance.

Technical report:

The report accounts for 40% of the project assessment and must contain five chapters addressing the following content areas (suggested):

- Fire sensing/extinguishing system (sensor/motor/circuit/ programming etc), 30%, by student 1
- Obstacle sensing system (sensor/circuit/programming etc), 30%, by student 2
- Behaviour control (BBC/FLC/FSM, codes etc), 30%, by student 3
- System integration (system configuration in SW and HW, MCU pin allocations and functions, flow charts, architecture, pseudo codes etc), 30%, by student 4
- **Project management, 10% in group**

Each chapter including Project Management is restricted to **two pages**, single line spacing and 11 font size of Arial. **Introduction, Conclusion and Appendix are not required and not marked if given.**

For each subsystem, write about “what”, “how” and “results”.

- “What” – what (sub)system you have worked on, including problem statement, scenarios, requirements, assumptions etc.
- “How” – how the (sub)system was designed, interfaced to MCU, driven by program, tested/calibrated, etc. How the entire system was integrated.
- “Results” – evidence of how the (sub)system works, including data, charts, diagrams, discussion of results etc.

For the project management chapter, write about the following:

- Have you applied any project management skills/knowledge learnt in *ENGGEN 303 Managing Projects and Innovation* into this project? If yes, what is it and how? If not, summarise the learnt skills/knowledge that may be applied to this project. (25%)
- Reflect how you have practised **Agile** product development technique (Lecture 2/Week 1) into this project. (25%)
- Describe any software (e.g., Jira, Microsoft Projects) used, or evaluated for potential use in this project. (25%)
- State who has contributed to which part of the project, and to what extent. (25%)

The report will be assessed according to the following rubrics:

- Technical contents, 70%, such as method, math, symbols, description, ...
- Presentation, 15%, such as figure, pseudo code, flowchart, diagram, ...
- Formatting/Language, 15%, such as grammar, font, logic, pagination, ...